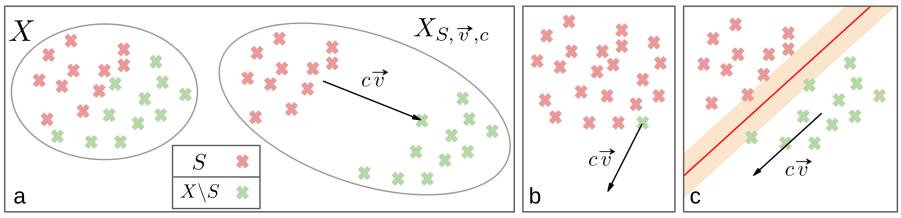

Figure 1: Visual representations of the transformations considered in the theoretical results. **a) Thm. 1:** The original set of neural representations X contains subsets S (red) and $X \backslash S$ (green). We can then build $X_{S,\vec{v},c}$ as a copy of X, where the points in $X \backslash S$ are translated a distance c in direction $\vec{v}$. The linear CKA value between X and $X_{S,\vec{v},c}$ is then computed. **b) Cor. 3:** X and $X_{S,\vec{v},c}$ differ by a single point, which has been translated by $c\vec{v}$ in $X_{S,\vec{v},c}$. **c) Cor. 4:** S and $X \backslash S$ are linearly separable (red line with orange margins), the transformation made to obtain $X_{S,\vec{v},c}$ preserves the linear separability of the data as well as the margins.

Corollary 2. *Thm. 1 holds even if S is taken such that $\rho = \frac{|S|}{|X|} \in (0.5, 1)$.*

Our main theoretical result is presented in Thm. 1 and Cor. 2, whose proofs are provided in the appendix along with additional details. These show that any set of internal neural representations X (e.g., from hidden layers of a network) can be manipulated with simple transformations (translations of a subset, see Fig. 1.a) to significantly reduce the CKA between the original and manipulated set. We note that our theoretical results are entirely class and direction agnostic (except for Cor. 4 which isn't direction agnostic), see the last paragraph of Sec. 4.2 for more details on this point.

Specifically, we consider a copy of X to which we apply a transformation where the representations of a subset $X \backslash S$ of the data is moved a distance c along direction $\vec{v}$, resulting in the modified representation set $X_{S,\vec{v},c}$. A closed form solution is found for the limit of the linear CKA value between X and $X_{S,\vec{v},c}$ as c tends to infinity. We note that up to orthogonal transformations (which CKA is invariant to) the transformation $X \mapsto X_{S,\vec{v},c}$ is not difficult to implement when transforming representations between hidden layers in neural networks. More importantly, it is also easy to eliminate or ignore in a single layer transformation, as long as the weight vectors associated with the neurons in the subsequent layer are orthogonal to $\vec{v}$. Therefore, our results show that from a theoretical perspective, CKA can easily provide misleading information by capturing representation differences introduced by a shift of the form $X \mapsto X_{S,\vec{v},c}$ (especially with high magnitude c), which would have no impact on network operations or their effective task-oriented data processing.

The terms in Eq. 2 can each be analyzed individually. $\Gamma(\rho)$ depends entirely on ρ, the proportion of points in $X_{S,\vec{v},c}$ that have not been translated i.e. that are exactly at the same place as in X. Its value is between 0 and 1 and it tends towards 0 for small sizes of S. The participation ratio, with values in $[1, p]$, is used as an effective dimensionality estimate for internal representations (Mingzhou Ding & Dennis Glanzman, 2011; Mazzucato et al., 2016; Litwin-Kumar et al., 2017). It has long been observed that the effective dimensionality of internal representations in neural networks is far smaller than the actual number of dimensions of the representation space (Farrell et al., 2019; Horoi et al., 2020). $\mathbb{E}_{x \in X}[\|x\|^2]$ and $\|\mathbb{E}_{x \in S}[x]\|^2$ are respectively the average squared norms of all representations in X and the squared norm of the mean of S, the subset of representations that are not being translated. Since most neural networks are trained using weight decay, the network parameters, and hence the resulting representations as well as these two quantities are biased towards small values in practice.

CKA sensitivity to outliers As mentioned in Sec. 2, it was recently found that the block structure in CKA heatmaps of high capacity models was caused by a few dominant data points that share similar characteristics (Nguyen et al., 2022). Other work has empirically highlighted CKA's sensitivity to directions of high variance, failing to detect important, function altering changes that occur in all but the top principal components (Ding et al., 2021). Cor. 3 provides a concrete explanation to these phenomena by treating the special case of Thm. 1 where only a single point, $\hat{x}$ is moved and thus has a different position in $X_{S,\vec{v},c}$ with respect to X, see Fig. 1.b for an illustration. We note that the term "subset translation" was coined by us and wasn't used in past work. However, all the papers referenced in this paragraph and later in this section present naturally occurring examples of subset translations in a set of representations relative to another, comparable set.

Corollary 3. *Thm. 1 holds in the special case where $S = \{\hat{x}\}$ is a single point, i.e. an outlier.*

Cor. 3 exposes a key weakness of linear CKA: its sensitivity to outliers. Consider two sets of representations that are identical in all aspects except for the fact that one of them contains an outlier, i.e. a representation further away from the others. Cor. 3 then states that as the difference

between the outlier's position in the two sets of representations becomes large the CKA value between the two sets drops dramatically, indicating high dissimilarity. Indeed, as previously noted, $\frac{\|\mathbb{E}_{x\in S}[x]\|^2}{\mathbb{E}_{x\in X}[\|x\|^2]}\sqrt{\dim_{PR}(X)}$ will be of relatively small value in practice so the whole expression in Eq. 2 will be dominated by $\Gamma(\rho) = \frac{\rho}{1-\rho}$. In the outlier case $\Gamma(\rho) \approx \rho = \frac{1}{\#\text{ representations}}$ which will be extremely small since for most modern deep learning datasets the number of examples in both the training and test sets is in the tens of thousands or more. This will drastically lower the CKA value between the two considered representations despite their obvious similarity.

CKA sensitivity to transformations preserving linear separability Classical machine learning theory highlights the importance of data separability and of margin size for predictive models generalization (Lee et al., 1995; Bartlett & Shawe-Taylor, 1999). Large margins, i.e. regions surrounding the separating hyperplane containing no data points, are associated with less overfitting, better generalization and greater robustness to outliers and to noise. The same concepts naturally arise in the study of ANNs with past work establishing that internal representations become almost perfectly linearly separable by the network's last layer (Zeiler & Fergus, 2014a; Oyallon, 2017; Jacobsen et al., 2018; Belilovsky et al., 2019; Davari & Belilovsky, 2021). Furthermore, the quality of the separability, the margin size and the decision boundary smoothness have all been linked to generalization in neural networks (Verma et al., 2019). Given the theoretical and practical importance of these concepts and their natural prevalence in deep learning models it is reasonable to assume that a meaningful way in which two sets of representations can be "similar" is if they are linearly separable by the same hyperplanes in representation space and if their margins are equally as large. This would suggest that the exact same linear classifier could accurately classify both sets of representations. Cor. 4 treats this exact scenario as a special case of Thm. 1, see Fig. 1.c for an illustration. If X contains two linear separable subsets, S and $X\backslash S$, we can create $X_{S,\vec{v},c}$ by translating one of the subsets in a direction that preserves the linear separability of the representations and the size of the margins while simultaneously decreasing the CKA between the original and the transformed representations, counterintuitively indicating a low similarity between representations.

Corollary 4. *Assume S and $X\backslash S$ are linearly separably i.e. $\exists w \in \mathbb{R}^p$, the separating hyperplane's normal vector, and $k \in \mathbb{R}$ such that for every representation $x \in X$ we have: $x \in S \Rightarrow \langle w, x\rangle \leq k$ and $x \in X\backslash S \Rightarrow \langle w, x\rangle > k$. We can then pick $\vec{v}$ such that S and $\{x + c\vec{v} : x \in X\backslash S\}$ are linearly separable by the exact same hyperplane and with the exact same margins as S and $X\backslash S$ for any value of $c \in \mathbb{R}_{\geq 0}$ and Thm. 1 still holds.*

Extensions to nonlinear CKA As previously noted in Sec. 2, given the popularity of linear CKA, it is outside the scope of our work to theoretically analyze nonlinear kernel CKA. However one can consider extending our theoretical results to the nonlinear CKA case with symmetric, positive definite kernels. Indeed we know from reproducing kernel Hilbert space (RKHS) theory that we can write such a kernel as an inner product in an implicit Hilbert space. While directly translating points in the representations space would likely not drive CKA values down as in the linear case, it would suffice to find/learn which transformations in representations space correspond to translations in the implicit Hilbert space. Our results should hold if we apply the found transformations, instead of translations, to a subset of the representations. Although practically harder to implement than simple translations, we hypothesize that it would be possible to learn such transformations.

4 Experiments and Results

In this section we will demonstrate several counterintuitive results, which illustrate cases where similarity measured by CKA is inconsistent with expectations (Sec 4.1) or can be manipulated without noticeably changing the functional behaviour of a model (Sec. 4.2 and 4.3). We emphasize that only Sec. 4.2 is directly tied to the theoretical results presented in the previous section. On the other hand, sections 4.1 and 4.3 discuss empirical results which are not explicitly related to the theoretical analysis.

4.1 CKA Early Layer Results

CKA values are often treated as a surrogate metric to measure the usefulness and similarity of a network's learned features when compared to another network (Ramasesh et al., 2021). In order to analyze this common assumption, we compare the features of: (1) a network trained to generalize on the CIFAR10 image classification task (Krizhevsky et al., 2009), (2) a network trained to "memorize" the CIFAR10 images (i.e. target labels are random), and (3) an untrained randomly initialized network (for network architecture and training details see the Appendix). As show in Fig. 2, early layers of these networks should have very similar representations given the high CKA values. Under the

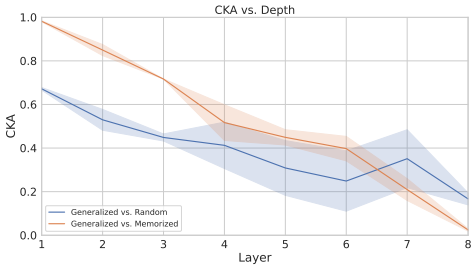

Figure 2: A layer-wise comparison based on the value of the CKA between a generalized, memorized, and randomly populated network. This comparison reveals that early layers of these networks achieve relatively high CKA values. Mean and standard deviation across 5 seeds are shown.

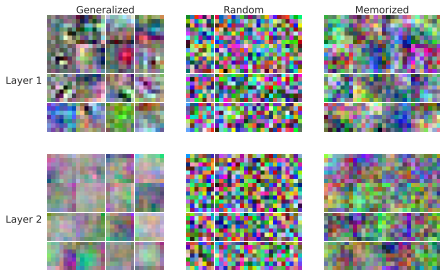

Figure 3: The convolution filters within the first two layers of a generalized, memorized, and a randomly initialized network elucidates that the features are (1) drastically different, and (2) not equally useful despite the CKA results in Fig. 2

previously presented assumption, one should therefore conclude that the learned features at these layers are relatively similar and equally valuable. However this is not the case, we can see in Fig. 3 that the convolution filters are drastically different across the three networks. Moreover, Fig. 3 elucidates that considerably high CKA similarity values for early layers, does not necessarily translate to more useful, or similar, captured features. In Fig. 16 of the Appendix we use linear classifying probes to quantify the "usefulness" of learned features and those results further support this claim.

4.2 Practical implications of theoretical results

Here we empirically test the behaviour of linear and RBF CKA in situations inspired by our theoretical analysis, first in an artificial setting, then in a more realistic one. We begin with artificially generated representations $X \in \mathbb{R}^{n \times d}$ to which we apply subset translations to obtain $Y \in \mathbb{R}^{n \times d}$, similar to what is described in Thm. 1. We generate X by sampling 10K points uniformly from the 1K-dimensional unit cube centered at the origin and 10K points from a similar cube centered at $(1.1, 0, 0, \ldots, 0)$, so the points from the two cubes are linearly separable along the first dimension. We translate the representations from the second cube in a random direction sampled from the d-dimensional ball and we plot the CKA values between X and Y as a function of the translation distance in Fig. 4a. This transformation entirely preserves the topological structure of the representations as well as their local geometry since the points sampled from each cube have not moved with respect to the other points sampled from the same cube and the two cubes are still separated, only the distance between them has been changed. Despite these multiple notions of "similarity" between X and Y being preserved, the CKA values quickly drop below 0.2 for both linear and RBF CKA. While our theoretical results (Thm. 1) predicted this drop for linear CKA, it seems that RBF CKA is also highly sensitive to translations of a subset of the representations. Furthermore, it is surprising to see that the drop in CKA value occurs even for relatively small translation distances. We note that RBF CKA with σ equal to 0.2 times the median distance between examples is unperturbed by the considered transformation. However, as we observe ourselves (RBF CKA experiments in the supplement) and as was found in the original CKA paper (see Table 2 of Kornblith et al. (2019)), RBF CKA with $\sigma = 0.2 \times$ median is significantly less informative than RBF CKA with higher values of σ. With small values of σ, RBF CKA only captures very local, possibly trivial, relationships.

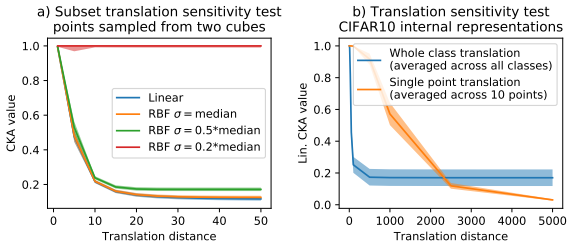

Figure 4: **a)** Linear and RBF CKA values between the artificial representations X and the subset translated version Y as a function of the translation distance. **b)** CKA value between a CNN's internal representations of the CIFAR10 training set and modified versions where either a class or a single point is translated as functions of the translation distance.

In a more realistic setting we test the practical implications of linear CKA sensitivity to outliers (see Cor. 3) and to transformations that preserve the linear separability of the data as well as the margins (see Cor. 4). We consider the 9 layers CNN presented in Sec. 6.1 of Kornblith et al. (2019) trained on CIFAR10. As argued in Sec. 2, when trained on classification tasks, ANNs tend to learn increasingly complex representations of the input data that can be almost perfectly linearly separated into classes